Q1 (2) - Single Correct

As, $\sum_{0 \le i<j<k<m \le n}\sum\sum\sum n = n \cdot \sum_{0 \le i<j<k<m \le n}\sum\sum\sum 1$

represents selection of four terms from $\{0, 1, 2 \ldots n\}$ i.e. $(n+1)$ terms

$\Rightarrow \quad {}^{n+1}C_4$

Thus, $\sum_{0 \le i<j<k<m \le n}\sum\sum\sum n = n \cdot {}^{n+1}C_4$

Hence, option (2) is correct.

Q2 (1) - Single Correct

Given, $n_1 = x_1x_2x_3$

and $n_2 = y_1y_2y_3$

n_1 and n_2 can be subtracted without borrowing at any stage, if $x_i \le y_i$

Value of x_3	Value of y_3
9	$0, 1, 2, \ldots 9$
8	$0, 1, 2, \ldots 8$
7	$0, 1, 2, \ldots 7$
6	$0, 1, 2, \ldots 6$
5	$0, 1, 2, \ldots 5$
4	$0, 1, 2, \ldots 4$
3	$0, 1, 2, 3$
2	$0, 1, 2$
1	$0, 1$
0	0

Thus, x_3 and y_3 can be selected collectively by $10 + 9 + 8 + \cdots + 1 = 55$ ways. Similarly, (x_2, y_2) can be selected in 55 ways. But pair (x_1y_1) can be selected in $1 + 2 + 3 + \cdots + 9 = 45$ ways as in this pair we cannot have zero. Thus, total number of ways in $45 \cdot (55)^2$.

Q3 (3) - Single Correct

All numbers $r!(r \ge 7)$ will be multiple of 7.

So, for remainder consider

$$\begin{aligned} S_6 &= 1+2+6+24+120+720 \\ &= 873 \end{aligned}$$

which leaves the remainder 5, when divided by 7

Q4 (4) - Single Correct

Sum of 7 digits is a multiple of 9. Sum of numbers $1, 2, 3, 4, 5$, 6,7,8,9 is 45. So, two left digits should also have sum of 9 . The pairs of left numbers are (1,8),(2,7),(3,6),(4,5) with each pair left numbers of 7 -digit number is 7!

So, with all 4 pairs, total number is $4 \times 7!$

Q5 (3) - Single Correct

Since, among 9! arrangements of digits 1,2,3,4,5,6,7,8,9

i.e. the number formed is of the form $x_1x_2x_3x_4x_5x_6x_7x_8x_9$ Taking any five consecutive positions product is divisible by 7 only, if $x_5 = 7$.

So, the required arrangement $= 8!$ Hence, option (c) is correct.

Q6 (3) - Single Correct

First we take one number as 2 and one as $(n-2)$ and put them in $m(m-1)$ ways. Now, remaining $(m-2)$ numbers can be anyone form $3, 4, \ldots (n-4), (n-3)$ which we can do in $(n-5)^{(m-2)}$

$\therefore$ Total number of ways $= m(m-1)(n-5)^{m-2}$

Q7 (2) - Single Correct

A rational number of the desired category is of the form

$2014x_1x_2\ldots x_k,$ where $1 \leq k \leq 9$

and $9 \geq x_1 \geq x_2 \geq \cdots \geq x_k \geq 1$. We can choose k digits out of 9 in ${}^3\mathrm{C}_k$ ways and arrange them in decreasing order in just one way. Thus, the desired number of rational number is

${}^9C_1 + {}^9C_2 + \cdots + {}^9C_9 = 2^9 - 1$

Q8 (1) - Single Correct

The number of ways of choosing first couple is $\left({ }^{15} C_1\right)\left({ }^{15} C_1\right)=(15)^2$

The, number of ways of choosing second couple is $\left({ }^{14} C_1\right)\left({ }^{14} C_1\right)=(14)^2$ and so on.

Thus, the number of ways of choosing the couples is $(15)^2+(14)^2+(13)^2+\cdots+2^2+1^2$

$=\frac{15 \times 16 \times(15 \times 2+1)}{6}=\frac{15 \times 16 \times 31}{6}\left[\because 1^2+2^2+\cdots+n^2=\frac{n(n+1)(2 n+1)}{6}\right]$

$=1240$

Q9 (2) - Single Correct

If n is odd, then $3^n=4 \lambda_1-1,5^n=4 \lambda_2+1$

$\Rightarrow \quad 2^n+3^n+5^n$ is divisible by 4, if $m \geq 2$.

Thus, $n=3,5,7,9, \ldots, 99$

i.e. n can take 49 different values.

If n is even, then $3^n=4 \lambda_1+1,5^n=4 \lambda_2+1$

$\Rightarrow 2^n+3^n+5^n$ is not divisible by 4

as $2^n+3^n+5^n$ will be in the form of $4 \lambda+2$

Then, total number of ways of selecting $' n'=49$.

Q10 (3) - Single Correct

Number of ways of making 30 in 7 shots is coefficient of x^{30} in

$\left(x^0+x^1+x^2+\cdots+x^5\right)^7$

$\Rightarrow$ Coefficient of x^{30} in $\left\{\left(1+n^2+n^3\right)+x^4(x+1)\right\}^7$

$\Rightarrow$ Coefficient of x^{30} in $\left\{x^{28}(x+1)^7+{ }^7 C_1 x^{24}(x+1)^6\left(1+x^2+x^3\right)\right.$

$\left.+{ }^7 C_2 x^{20}(x+1)^5\left(x^3+x+1\right)^2+\cdots\right\}$

$\Rightarrow \quad{ }^7 C_5+{ }^7 C_1\left({ }^6 C_3+{ }^6 C_2+{ }^6 C_0\right)+{ }^7 C_2\left({ }^5 C_1+2\right)$

$\Rightarrow \quad 21+252+147=420$

Q11 (2) - Single Correct

Here, $8!=2^7 \cdot 3^2 \cdot 5^1 \cdot 7^1$

Obviousiy, the factors are not multiple of either 2 or 3. So, the factors may be 1,5,7,35 of which 5 and 35 are of the form $3m+2$

So, the sum $=40$

Q12 (2) - Single Correct

Any number will be of the form $4t+1$, if it does not contain any power of 2, contain even number of odds of the form 4k $+3$ and any number of odds of the form $4k+1$.

So, total factors $=1\times3\times4=12$

But, one will be 1 also, So, required number $=12-1=11$

Alternate Method

Odd numbers of the type $4t+1$ or $4t+3$ and they are symmetrical. $\Rightarrow$ Number of odd numbers

$=(1+5)(1+3)=24$

(any number of 5's and 3's and no 2's)

$\Rightarrow$ Required number $=12$, but 1 is included

$\Rightarrow$ Required number of numbers $=12-1=11$

Q13 (4) - Single Correct

Triplet $x=y<z, x<y<z, y<x<z$ can be choosen in ${}^{n+1}C_2, {}^{n+1}C_3, {}^{n+1}C_3$ ways.

$\therefore\ {}^{n+1}C_2+2\cdot{}^{n+1}C_3={}^{n+2}C_3+{}^{n+1}C_3 \quad \left[\because {}^{n}C_r+{}^{n}C_{r-1}={}^{n+1}C_r\right]$

Q14 (4) - Single Correct

In first round all the integers, which leaves the remainder 1, when divided by 15 will be marked.

Last number of this category is 991.

Next number to be marked in $(991+15-1000)=6$ Again, second round of integers, which leaves the remainder 6, when divided by 15, will be marked.

Last number of this category is 996. Next number to be marked is $(996+15-1000)=11$

Thus, thrid round of integers which leave the remainder 11 when divided by 15, will be marked. Least number of this category is 986.

Next number to be marked is $(986+15-1000)=1$ which is already been marked.

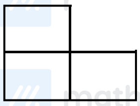
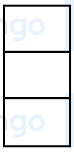
Hence, the number of the form $15\lambda+1, 15\lambda+6, 15\lambda+11$, are marked. All these numbers can also be put in the form of $5\lambda_1 + 1\ (\lambda_1 \geq 0)$

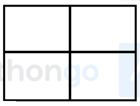
Hence, total number of marked numbers $= 1 + \left[\frac{999}{5}\right] = 200$

$\therefore$ Unmarked numbers $= 800$

Q15 (2) - Single Correct

Either, we have to choose

 or 

(i) Every square of (2 by 2) 

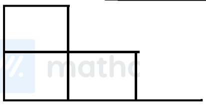
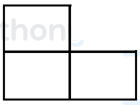
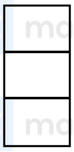
will contribute four  shaped

figure by removing anyone square out of it.

Number of ways to choose these square $= 7 \times 7 = 49$

So, total  shaped figure $= 49 \times 4 = 196$

(ii) In every line, it is possible to have 6  shaped figure.

So, total number of such figure $= 6 \times 8 \times 2 = 96$

$\therefore$ Total number of cases $= 196 + 96 = 292$

Q16 (3) - Single Correct

First, we have to find $f(6)$

$$\begin{aligned} \therefore \quad 6 &= 0(2) + 6(1) = 1(2) + 4(1) \\ &= 2(2) + 2(1) = 3(2) + 0(1) \end{aligned}$$

Number of 2's	Number of 1's	Number of permutation
0	6	1
1	4	$\frac{5!}{4!}=5$
2	2	$\frac{4!}{2!2!}=6$
3	0	$\frac{3!}{3!}=1$
	Total	13

$\therefore \quad f(6)=13$

Now $\quad f(f(6))=f(13)$

Q17 (2) - Single Correct

Given digits $8,7,6,4,2,x,y$

We know that, any number is divisible by 3, if sum of digit is divisible by 3

i.e. $8+7+6+4+2+x+y$

$=(x+y+27)$ is divisible by 3

SO, x and y can take the values from 0,1,3,5,9 Possible pairs are (5,1)(3,0)(9,0)(9,3) and (1,5)(0,3)(0,9)(3,9)

Hence, total number of ordered pair (x,y) is 8.

Q18 (2) - Single Correct

Let A,B,C want to remain together and D,E do not want to remain together.

In	Out	Number of Selection
A,B,C,D	E	${}^5C_1=5$
A,B,C,E	D	${}^5C_1=5$
D	A,B,C,E	${}^5C_4=5$
E	A,B,C,D	${}^5C_4=5$
	Total	20

Q19 (3) - Single Correct

We have, $1000=2^3\cdot 5^3$

$2000=2^4\cdot 5^3$

So, $a=2^A\cdot 5^R, b=2^B\cdot 5^S, c=2^C\cdot 5^T$

$\max(A, B)=3, \max(A, C)=\max(B, C)=4$

$\max(R, S)=\max(R, T)=\max(S, T)=3$

There are 10 ways of choosing R, S, T. 1 with all 3,3 with one 2,3 with one 1,3 with one $0, C$ must be 4 There are 7 ways of choosing A, B. 1 with both 3,3 with $A=3, B$ not, 3 with $B=3$ and A not.

Thus, 7 ways of choosing A, B, C have $7 \cdot 10=70$ ways.

Alternate Method

$$\begin{aligned}\text{LCM of }(a, b)&=1000=2^3 \cdot 5^3=2^{\alpha_1} \cdot 5^{\alpha_2}\\ \text{LCM of }(b, c)&=2000=2^4 \cdot 5^3=2^{\beta_1} \cdot 5^{\beta_2}\\ \text{LCM of }(a, c)&=2000=2^4 \cdot 5^3=2^{\gamma_1} \cdot 5^{\gamma_2}\end{aligned}$$

$\operatorname{Max}(\alpha_1, \beta_1)=4, \max(\beta_1, \gamma_1)=4, \max(\gamma_1, \alpha_1)=4$

$\operatorname{Max}(\alpha_2, \beta_2)=3, \max(\beta_2, \gamma_2)=3, \max(\gamma_2, \alpha_2)=3$

If $\alpha_1>\beta_1 \Rightarrow \alpha_1=3, \beta_1<3$ and $\gamma_1=4$

$\therefore \alpha_1=3, \gamma_1=4$ and $\beta_1=0,1,2 \quad$ [so, 3 possibilites]

If $\alpha_1<\beta_1 \Rightarrow \beta_1=3, \alpha_1<3$ and $\gamma_1=4$

$\therefore \alpha_1=0,1,2, \beta_1=3, \gamma_1=4 \quad$ [3 possibilities]

If $\alpha_1=\beta_1=3$ and $\gamma_1=4 \quad$ [1 possibilities]

$\therefore$ Total of 7 possibilities for $\alpha_1, \beta_1, \gamma_1$ If $\alpha_2>\beta_2 \Rightarrow \alpha_2=3$

$\beta_2<3$ and $\gamma_2=3 \quad$ [3 possibilities] If $\alpha_2<\beta_2 \Rightarrow \beta_2=3, \alpha_2<3$ and $\gamma_2=3 \quad$ [3 possibilities]

If $\alpha_2=\beta_2=3 \Rightarrow \gamma_2 \leq 3 \quad$ [4 possibilities]

$\therefore$ Total 10 possibilities for $\alpha_2, \beta_2, \gamma_2$.

Hence, total triplets $=7 \times 10=70$

Q20 (3) - Single Correct

Total number of ways to place 11 identical balls $={ }^{13}\mathrm{C}_2$

Total number of ways to place the balls so that the first box will have more balls than the second and third boxes combined $={ }^7\mathrm{C}_2$

$\therefore$ Required cases $={ }^{13}\mathrm{C}_2-3 \cdot{ }^7\mathrm{C}_2$

Q21 (3,4) - Multiple Correct

If a, b and c are in AP, then a and c both are odd or both are even.

Case I When n is even

The number of ways of selection of two even numbers a and c is ${}^{n/2}C_2$. Number of ways of selection of two odd numbers is ${}^{n/2}C_2$

Hence, the number of A.P is $2 \cdot \cdot {}^{n/2}C_2 = 2 \cdot \frac{\frac{n}{2}\left(\frac{n}{2}-1\right)}{2} = \frac{n(n-2)}{4}$

Case II When n is odd

The number of ways of selection of two odd numbers a and c is ${}^{\frac{n+1}{2}}C_2$. The number of ways of selection of two even numbers a and c is ${}^{\frac{n-1}{2}}C_2$.

Hence, the number of A.P's is

$$
\begin{aligned}
{}^{\frac{n+1}{2}}C_2 + {}^{\frac{n-1}{2}}C_2 &= \frac{\left(\frac{n+1}{2}\right)\left(\frac{n+1}{2}-1\right)}{2} + \frac{\left(\frac{n-1}{2}\right)\left(\frac{n-1}{2}-1\right)}{2} \\
&= \frac{1}{8}(n-1)(n+1)+(n-3) = \frac{(n-1)^2}{4}
\end{aligned}
$$

Q22 (2,4) - Multiple Correct

The combinatorial coefficient ${}^{n-1}\mathrm{C}_p$ denotes the number of ways in which p different things can be selected out of n different things, if a particular thing is always excluded.

(d) First, we arranged $(n-2)$ white balls in a line. Since, balls are identical, then only one such arrangement is possible. Now, there are $(n-2+1)$ i.e. $(n-1)$ places, where we can arrange

p black balls.

$\therefore$ Number of arrangement of p identical black balls in $(n-1)$ places $= {}^{n-1}C_p$

Thus, required number of ways $= 1 \cdot {}^{n-1}C_p = {}^{n-1}C_p$

Q23 (2,3,4) - Multiple Correct

We know that, distribution of n distinct objects in to I different boxes, if empty boxes are not allowed or in each box atleast one object is put $(n > r)$

The number of ways $= r^n - {}^rC_1(r-1)^n + {}^rC_2(r-2)^n + \ldots (-1)^{r-1}\,{}^rC_{r-1}1.$

Hence, $n = 5, r = 3$

$\therefore$ Number of ways in which the different books be distributed in 3 persons, so each gets atleast one is

$(3)^5 - {}^3C_1(2)^5 + {}^3C_2 = 243 - 3 \times 32 + 3 = 150$

(2) Number of parallelogram formed by one set of 6 parallel lines and other set of 5 parllel lines

$= {}^6C_2 \times {}^5C_2 = 15 \times 10 = 150$

(3) 5 different toys to be distributed among 3 children, so each

gets atleast one

$= (3)^5 - {}^3C_1(2)^5 + {}^3C_2$
$= 243 - 3 \times 32 + 3 = 150$

(4) 3 Mathematics professors are assigned five different lectures,

so that each prolessor get at least one lecture.

$= (3)^3 - {}^3C_1(2)^5 + {}^3C_2 - 1$
$= 243 - 3 \times 32 + 3 = 150$

Q24 (3) - Matrix Match (Only one option is correct)

Required number of ways $= {}^9C_3 = \frac{9!C_0}{3!6!} = 84$

$\therefore$ (i) $\rightarrow$ (t)

(ii) Total number of triangles $= {}^9C_3 = \frac{9!}{3!6!} = 84$

There are 4 set of 3 collinear points which do not make triangles.

$\therefore T = 84 - 4 = 80$

Total number of straight lines $= {}^9C_2 = \frac{9!}{2!7!} = 36$

$\therefore$ There are 8 straight lines which are sides of regular hexagon.

$\therefore \quad S = 36 - 8 = 28$

Now $\quad T - S = 80 - 28 = 52$

$\therefore$ (ii) $\rightarrow (r)$

(iii) All are get their own mobile $= {}^5C_5 = 1$ ways One of them do not get their own mobile $= {}^5C_4 = 5$

Two of them do not get their own

$= {}^5C_3 \cdot \left(1 - \frac{1}{1!} + \frac{1}{2!}\right) = 10 \times \frac{1}{2} = 5$

$\therefore$ Required number of ways $= 1 + 5 + 5 = 11$

$\therefore$ (iii) $\rightarrow$ (p)

(vi) Here, $12 = 2^2 \times 3$

Hence, groups of 4 that work out

$1,1,6,2=\frac{4!}{2!}=12$

and $1,2,3,4=\frac{4!}{2!}=12$

$2,2,3,1=\frac{4!}{2!}=12$

Total=36

$\therefore$ (iv) $\rightarrow(q)$

(v) First we select 2 men out of 75 in ${}^{5}\mathrm{C}_{2}$ ways. Now, we exude the wives of these two selected men and so select 2 ladies from remaining 3 ladies in ${}^{3}\mathrm{C}_{2}$ ways. Let A and B be two men and C, D are two ladies playing in

one set then, we hare

(a) A and C playing against B, D

(b) A and D playing against B, C Total number of ways $=2\times{}^{5}\mathrm{C}_{2}\times{}^{3}\mathrm{C}_{2}=2\frac{5!}{2!3!}\times\frac{3!}{2!1!}=60$

$\therefore(v)\rightarrow(s)$

Q25 (2) - Paragraph

$a_n+b_n+c_n$

$b_n=a_{n-1},\quad c_n=a_{n-2}$

$\Rightarrow a_n=a_{n-1}+a_{n-2}$

As, $a_1=1, a_2=2, a_3=3, a_4=5, a_5=8$

$b_6=8$

Q26 (1) - Paragraph

$As, a_n=a_{n-1}+a_{n-2}$, for $n=17$

$\Rightarrow a_{17}=a_{16}+a_{15}$

Q27 (4) - Paragraph

Explanation (For Question 27, 28, 29)

Let $N=x_1, x_2, x_3, \cdots, x_{10}$ be one of the such numbers. $\because\Sigma x_i=45$, hence N is divisible by 9, but 11111 is not divisible by

9. Hence, N should be divisible by $11111 \times 9 = 99999$. Now,

$x_1 x_2 x_3 \cdots x_9 x_{10} = x_1 x_2 x_3 x_4 x_5 \times 10^5 + x_6 x_7 x_8 x_9 x_{10}$

$= x_1 x_2 x_3 x_4 x_5 \times (99999) + x_1 x_2 x_3 x_4 x_5 + x_6 x_7 x_8 x_9 x_{10}$

Now, $x_1 x_2 x_3 x_4 x_5 < 99999$

and $x_6 x_7 x_8 x_9 x_{10} < 99999$

$\therefore \quad x_1 x_2 x_3 x_4 x_5 + x_6 x_7 x_8 x_9 x_{10} < 2(99999)$

$\Rightarrow \quad x_1 x_2 x_3 x_4 x_5 + x_6 x_7 x_8 x_9 x_{10} = 99999$

$\Rightarrow \quad x_1 + x_6 = x_2 + x_7 = x_3 + x_8 = x_4 + x_9 = x_5 + x_{10} = 9$

Smallest number $= 1023489765$.

Q28 (1) - Paragraph

Biggest number $= 9876501234$.

Q29 (3) - Paragraph

Total ways $= 9 \times 8 \times 6 \times 4 \times 2 \times 1 \times 1 \times 1 \times 1 \times 1 = 3456$

Q30 (2) - Paragraph

$x_R + x_G + x_W + x_y + x_B = 12, x_R, x_G, x_W, x_y, x_B \geq 0$

Total number of non-negative integral solutions of above equation $= {}^{12+5-1}C_{5-1} = {}^{16}C_4$

Q31 (2) - Paragraph

$x_R + x_G + x_W + x_y + x_B = 12$, where $x_R, x_G, x_W, x_y, x_B \geq 1$

$\therefore$ Number of solutions $= {}^{7+5-1}C_{5-1} = {}^{11}C_4$

Q32 (3) - Paragraph

$x_R + x_G + x_W + x_y + x_B = 16$

Where, $x_R, x_G, x_W, x_y, x_B \geq 2$

$\therefore$ Number of solutions $= {}^{6+5-1}C_{5-1} = {}^{10}C_4$

Q33 (1) - Paragraph

$x_1 < x_2 < x_3 < x_4 < x_5 < x_6 \Rightarrow {}^9C_6$ ways

Q34 (2) - Paragraph

$x_1 < x_2 < x_3 = x_4 < x_5 < x_6 \Rightarrow {}^9C_4$ ways

Q35 (3) - Paragraph

$x_1 < x_2 < x_3 \leq x_4 < x_5 = x_6$
$\Rightarrow \quad {}^9C_6 + {}^9C_4 = {}^9C_3 + {}^9C_4 = {}^{10}C_4$ ways

Q36 (4) - Paragraph

Required number of ways $= (3)^5 - {}^3C_1(3-1)^5 - {}^3C_2(1)^5$

$= 243 - 3 \times 32 + 3 = 150$

Q37 (2) - Paragraph

When balls are identical, but boxes are different

Box I	Box II	Box III	Number of ways
1	2	2	1
1	1	3	1
1	3	1	1
2	1	2	1
3	1	1	1
2	2	1	1
		Total	6 ways

Alternate Method $x_1 + x_2 + x_3 = 5, x_1 \geq 1, x_2 \geq 1, x_3 \geq 1$

$\Rightarrow \quad k + 1 + m = 2, k \geq 0, 1 \geq 0, m \geq 0$

Number of solutions $= {}^{2+3-1}C_{3-1} = {}^4C_2 = 6.$

Q38 (1) - Paragraph

Here, two cases arises.

Case I $(2,2,1) \Rightarrow \frac{{}^5C_2 \cdot {}^3C_2 \cdot {}^1C_1}{2!} = 15$

Case II $(1,1,3) \Rightarrow \frac{{}^5C_1 \cdot {}^4C_1 \cdot {}^3C_3}{2!} = 10$

$\therefore$ Total number of ways $= 15 + 10 = 25$

Q39 (2) - Paragraph

Let's count the Δ's right away

FS	SS	TS
1	1	1
2	2	1, 2, 3
3	3	1, 2, 3, 4
4	4	1, 2, 3, 4

FS $\rightarrow$ First side SS $\rightarrow$ Second side

TS $\rightarrow$ Third side

$\therefore$ Total number of ways $= 1 + 3 + 4 + 4 = 12$

Q40 (3) - Paragraph

The value of equal sides can vary from 1 to 2P

Hence, third side $< 2, < 4, < 6, \ldots, < 4P$.

FS	SS	TS
1	1	1
2	2	1, 2, 3
3	3	1, 2, 3, 4, 5
4	4	1, 2, 3, 4, 5, 6, 7
:	:	:
p	p	$1, 2, \ldots, (2p-1)$
$p+1$	$p+1$	$1, 2, 3, \ldots 2p$
:	:	:
$2p$	$2p$	$1, 2, 3, \ldots, 2p$

$\therefore$ Number of $\Delta's = \{1 + 3 + 5 + 7 + \cdots + (2p-1)\} + \{2p + 2p + \cdots + p \text{ times}\}$

$= p^2 + 2p^2 = 3p^2$

Q41 (8) - Integer Type

Here, $P_n = {}^{n-2}C_3$ and $P_{n+1} = {}^{n-1}C_3$

Hence, ${}^{n-1}C_3 - {}^{n-2}C_3 = 15$

or ${}^{n-2}C_3 + {}^{n-2}C_2 - {}^{n-2}C_3 = 15$

or ${}^{n-2}C_2 = 15$

$\Rightarrow \quad \frac{(n-2)!}{2!(n-4)!} = 15$

$\Rightarrow \quad \frac{(n-2)(n-3)}{2} = 15$

$\Rightarrow \quad (n-2)(n-3) = 30$

$\Rightarrow \quad (n-2)(n-3) = 6 \times 5$

$\Rightarrow \quad (n-2)(n-3) = (8-2)(8-3)$

$\therefore \quad n = 8$

Q42 (1) - Integer Type

${}^{2002}C_{1001} = \frac{(2002)!}{(1001)!(1001)!}$

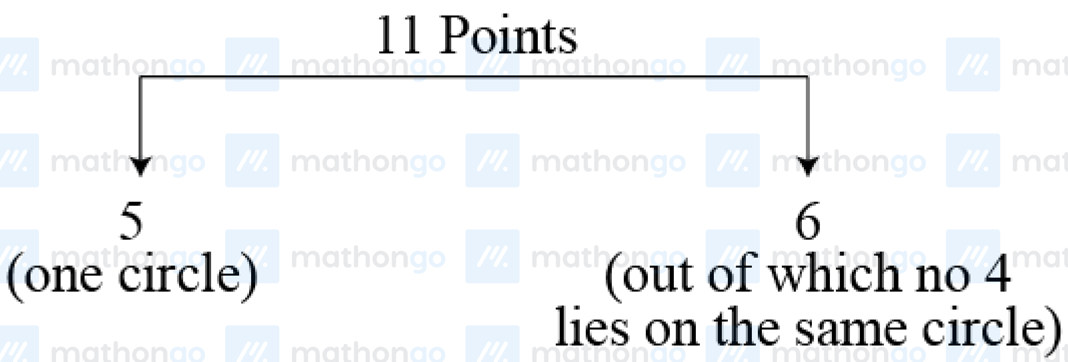
Number of zero's in (2002) ! are,

$= 400 + 80 + 16 + 3 = 499$

Number of zero's in $(1001!)^2 = 2(200 + 40 + 8 + 1) = 498$

Hence, number of zero's in $\frac{(2002)!}{(1001!)^2} = 1$

Q43 (3) - Integer Type



$\Rightarrow$ maximum number of circles

$$= {}^5C_2 \cdot {}^6C_1 + {}^6C_2 \cdot {}^5C_1 + {}^6C_3 + 1 = 156 = 52(3)$$

$\therefore \quad k = 3$

Alternate Method Maximum number of circles

$= {}^{11}C_3 - {}^5C_3 + 1$

Q44 (6) - Integer Type

We have, 2 identical white balls (W), 3 identical red balls (R) and 4 green balls of different shades (G_1, G_2, G_3, G_4). Let us first find the number of ways in which they can be arranged in a row, so that no ball is separated from the balls of the same colour.

i.e. WW RRR $\quad G_1G_2G_3G_4$

$\therefore$ Number of ways $= 3! \times 4!$

Also, total number of ways of arranging these balls in a row $= \frac{9!}{2!3!}$

$\therefore$ Required number of ways $= \frac{9!}{2!3!} - 3! \times 4!$

$$= \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4}{2} - 3! \times 4!$$
$$= 6(7 \times 6 \times 5 \times 4 \times 3 \times 2) - 6 \times 4!$$
$$= 6(7! - 4!)$$
$\therefore\ k = 6$

Q45 (9) - Integer Type

Number of dearrangements in such problems is given by

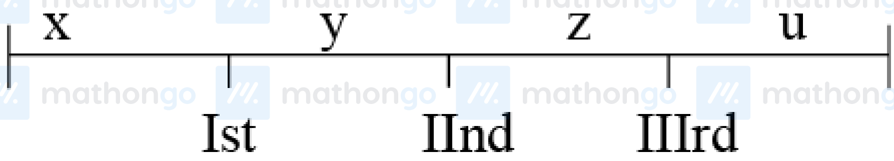
$n!\left\{1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots + (-1)^n \frac{1}{n!}\right\}$

Hence, the required number of dearrangements is

$4!\left\{\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right\} = 12 - 4 + 1 = 9$

Q46 (3) - Integer Type

Let x be the number of books to the left of the first book chosen, y be the number of books between the first and second, z be the number of books between the second and third and u be the number of books to the right of the third book



Then, $x, u \geq 0, y, z \geq 1$ and $x + y + z + u = 97$

Let $y_1 = y - 1$ and $z_1 = z - 1$. Then, $y_1 \geq 0, z_1 \geq 0$ such that

$x + y_1 + z_1 + u = 95$ ...(i)

$\therefore$ Total number of ways in which 3 books can be selected = Number of non-negative integral solution of (i)

$= {}^{95+4-1}C_{4-1} = {}^{98}C_3$

$\therefore k = 3$

Q47 (9) - Integer Type

Total number of digits $= 9$

Now, we can select 2 places for the digit 1 and 2 in ${}^{9}\mathrm{C}_2$ ways, from the remaining 7 places select any two places for 3 and 4 in ${}^{7}\mathrm{C}_2$ ways and from the remaining 5 places select any two for 5 and 6 in ${}^{5}\mathrm{C}_2$ ways.

Now, we left with 3 digits 7,8,9 and 3 places that can be filled in $3!$ ways.

$\therefore$ Total number of ways $= {}^{9}C_2 \cdot {}^{7}C_2 \cdot {}^{5}C_2 \cdot 3! = 9 \cdot 7!$

$\therefore k = 9$

Q48 (3) - Integer Type

Here, we have to find the number of different natural numbers which are less than 2×10^8 and which can be written by means of

digits 1 or 2 only. Clearly, the required numbers are $1, 2, 11, 12, 21, 22, \ldots, 122222222$.

Now, let us calculate how many numbers are these. There are 2 one digit such numbers. There are 2^2 two digit such numbers. And so on.

There are 2^8 eight digit such numbers. All the nine-digit numbers beginning with 1 and written by means of 1 or 2 are smaller than 2×10^8 and these are 2^8 in numbers.

Hence, required number of numbers

$= 2 + 2^2 + 2^3 + \cdots + 2^8 + 2^8$

$= \frac{2(2^8-1)}{2-1} + 2^8 = 2^9 - 2 + 2^8$

$= 2 \cdot 2^8 + 2^8 - 2 = 2^8(2+1) - 2 = 3 \cdot 2^8 - 2$

$\therefore k = 3$

Q49 (9) - Integer Type

Given, 5 Indian and 5 American couples meet at party and shake hand.

$\therefore$ Total number of possible hand shakes $= {}^{20}\mathrm{C}_2$

Number of hand shakes, when wife shake hand with her own husband $= 10$(5 Indian and 5 American)

Number of hand shakes when Indian wife shakes hand with a male $= {}^{5}C_1 \times {}^{10}C_1 = 50$ (it include the case where the Indian wife shake hand with her own husband)

$\therefore$ Total number of hand shakes that take place in the party

$= {}^{20}C_2 - 5 - 50 = 135 = 15 \times 9$

$\therefore k = 9$

Q50 (8) - Integer Type

There are 3 choices for the first of n -letter and two choices for each subsequent letters. Hence, using fundamental principle, number of good words

$= 3 \cdot 2^{n-1} = 384$

$= 2^{n-1} = 128 = 2^7$

$\therefore n = 8$